Lab Procedure
Passive Infrared

Introduction

Ensure the following:

1. You have reviewed the Application Guide – Passive Infrared

2. Make sure the **Mechatronic Sensors Trainer** is out of its case; it is powered using the provided USB cable connected to the PC.

    a. The LED next to the USB C port will stay white after the touch screen starts loading.

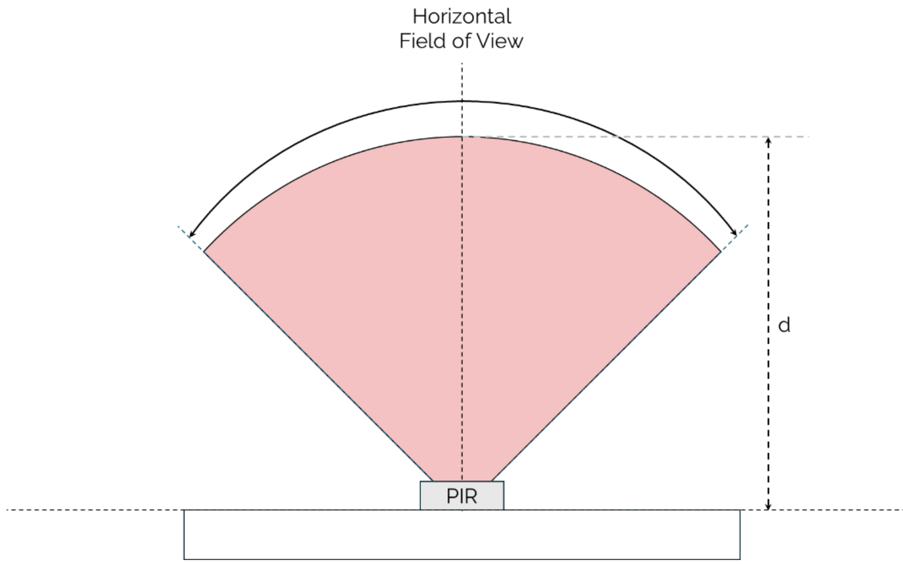
    b. The load screen with the Quanser Logo should disappear after a few seconds and you should see some evenly spaced crosses on a black background on the touch screen.

Exploring Passive Infrared Sensors

1. Launch Visual Studio Code or your preferred Python IDE. If using Visual Studio Code, click on **File** > **Open Folder**…, and browse to the directory containing the python files and lab procedure for the Passive Infrared lab (…/sensors_trainer/1_fundamentals /passive_infrared/hardware/python). Open this directory.

2. Print out a protractor on a piece of paper. Within the (…/sensors_trainer/1_fundamentals /passive_infrared) directory an example printable protractor is provided. Ensure the 90° mark is aligned with the Passive Infrared sensor.

3. For the first test we will estimate a field of view approximated as:



4. Open the provided Passive Infrared Datasheet located in the passive infrared folder (…/sensors_trainer/1_fundamentals/passive_infrared) and verify the viewing angle for a passive infrared sensor with a natural lens cover.

5. Open passive_infrared.py.

6. Click to run the experiment.

7. Make a note of the voltage reading at ambient temperature.

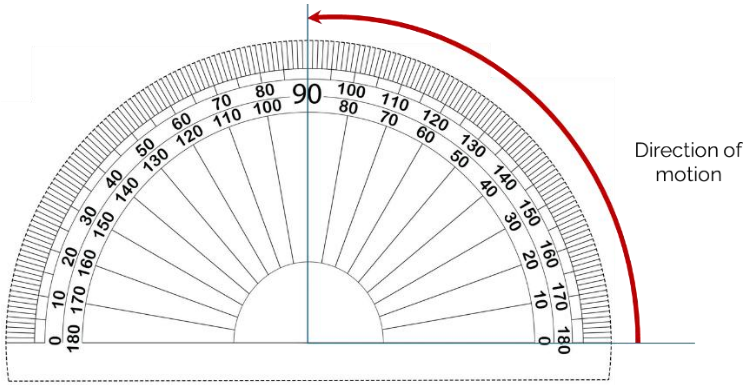
8. Based on the 018_passive_infrared concept review, do you expect the field of view will be affected by small/large temperature changes? Will the distance of the object also affect the field of view?

9. Grab a warm object such that there is a delta in temperature between ambient temperature and the object itself.

10. Re-run passive_infrared.py and monitor the voltage reading.

11. Being careful not to place your hand near or in front of the Passive Infrared Sensor:

    a. Place the object at the 0 ° mark (on the right side of the protractor), roughly 0.5m away from the sensor board.



    b. Slowly move the object towards to the 90° mark maintaining the 0.5m distance and while monitoring the voltage measurement from the Passive Infrared sensor.

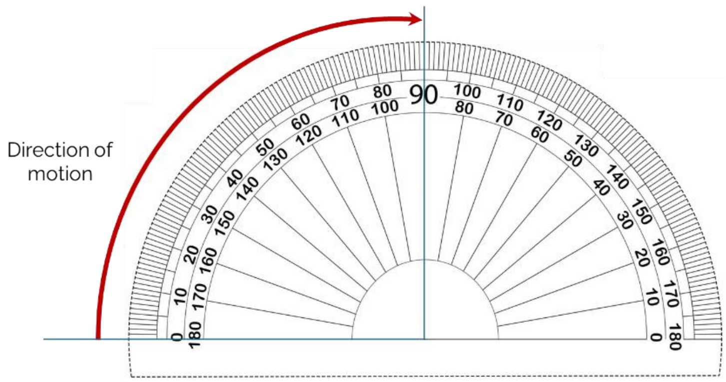
12. When a voltage spike is observed:

    a. stop moving the object,

    b. make a note of the angle at which the voltage spike occurred

    c. remove the object.

13. Redo the previous 2 steps except this time place the object at the 0° mark on the left side and bring it towards to the 90° mark.



14. What do you notice about the measured field of view compared to the field of view shown in the datasheet?

15. Once the field of view extremes have been identified, place the object 1.5m away from the sensor at the 0° mark.

16. Re-run passive_infrared.py and monitor the voltage reading.

17. Slowly move the object closer to the **Mechatronics Sensors Trainer**.
18. When a voltage spike is measured:
    a. Stop moving the object,
    b. Make a note of distance and voltage values.
19. Based on the tests performed, are Passive Infrared Sensors good for measuring distance, motion, both? Why?
20. Stop, save and close your program.
21. Power OFF the Mechatronic Sensors Trainer by disconnecting its USB C cable, disconnect any external sensors and pack them along the base and the USB cables in the provided case.